

SEN22106E Computer Programming II Laboratory Midterm Project Report

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Date: 16.05.2026

Project: Rent Payment Tracking

Rent Payment Tracking

1. The Main Purpose of the Program:

The Rent Payment Tracking program is designed to manage tenants and their payments accurately by using different layers: User Interface (UI) layer and the Object layer.

2. Objects and Relationships of the Program (Object Layer)

2.1. User:

Represents the authorized person accessing the system via specified username and password. It stores the username and password required to log in.

2.2. Tenant:

This object stores information about each renter such as name, apartmentNo and monthly rent amount they have to pay.

2.3. Payment:

A major object used to track money received from tenants. It includes apartment number, amount and date of payment.

2.4. OnTimePayment:

The first subclass that extends payment class (inheritance). It handles standard payments.

2.5. LatePayment:

The second subclass that extends payment class (inheritance). LatePayment includes a rule to add extra charge for delays.

2.6. ApartmentManager:

This object is the logic controller. It stores all Tenants and Payment of two types in ArrayLists and performs operations and some calculations like debts of tenants.

3. Frames and User Interface (UI Layer):

3.1. Login Screen:

This is the entry point of program. It checks user's username and password and decides the access to the main menu. For security demonstration, the username and password are displayed on the Main Menu as per project requirements

3.2. Main menu:

It contains a central ApartmentManager object. It provides buttons to navigate to different sections: Tenant Management, Payment Operations, and Financial Reports.

3.3. Tenants Management Screen:

To manage the list of people living in the apartments. Uses JTextFields for input and JTable to show current tenants. User can add, update and delete tenants' information. It communicates with ApartmentManager to keep the data consistent.

3.4. Payment Screen:

To record rent Payments. Use a JComboBox to select a tenant and a jtable to display a payment history. It distinguishes between On-Time and Late payments. When a payment is recorded, it calculates the final amount based on the rules defined in the object layer.

3.5. Report Screen:

Provides a financial overview of the property. Uses a JTable to list all transactions and JLabel components to display calculated totals. It calculates the Total Income from all payments and the Average Rent of all active tenants.

4. Complex Rules:

1. Each apartment can only have one tenant.
2. Debt based deletion restriction; if the tenants have a debt the delete operation will not be applied.
3. A tenant cannot pay more than their monthly rent for a single month.
4. If a payment is marked as "Late," then a %2 penalty fee is automatically added to amount.
5. Payments can be made in installments within the same month, provided that the total amount does not exceed the rent.

5. Mathematical Operations:

- 1.Total income calculation.
2. Average rent calculation.
3. Debt calculations.

